



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SECURE ONLINE VOTING SYSTEM USING CRYPTOGRAPHY

CSA5115-CRYPTOGRAPHY NETWORK SECURITY

PRESENTED BY

M. NISHANTH (192511154)

A Meghanadh (192372281)

N Charan Sai (192372003)

Ch.Mahi Vardhan Reddy (192411210)

UNDER THE GUIDANCE OF

Dr. T. SANGEETHA



INTRODUCTION



- Online voting makes the voting process easy and fast.
- Security and privacy are important in online voting.
- Votes must be protected from unauthorized access and tampering.
- Our project provides a **Secure Online Voting System Using Cryptography**.
- RSA encryption is used to secure the votes.
- Digital signatures verify the authenticity of votes.
- The system securely counts votes and generates results.



PROBLEM STATEMENT



- Online voting systems can face security and privacy issues.
- Unauthorized users may attempt to access the voting system.
- Votes can be modified or tampered with during transmission or storage.
- Duplicate voting may affect the accuracy of election results.
- Voter identity and vote confidentiality must be protected.
- A secure system is needed to ensure safe and reliable online voting.





OBJECTIVE



- To develop a secure online voting system.
- To register and authenticate eligible voters.
- To protect votes using RSA encryption.
- To verify vote authenticity using digital signatures.
- To prevent unauthorized and duplicate voting.
- To securely count votes and generate accurate results.



SYSTEM OVERVIEW



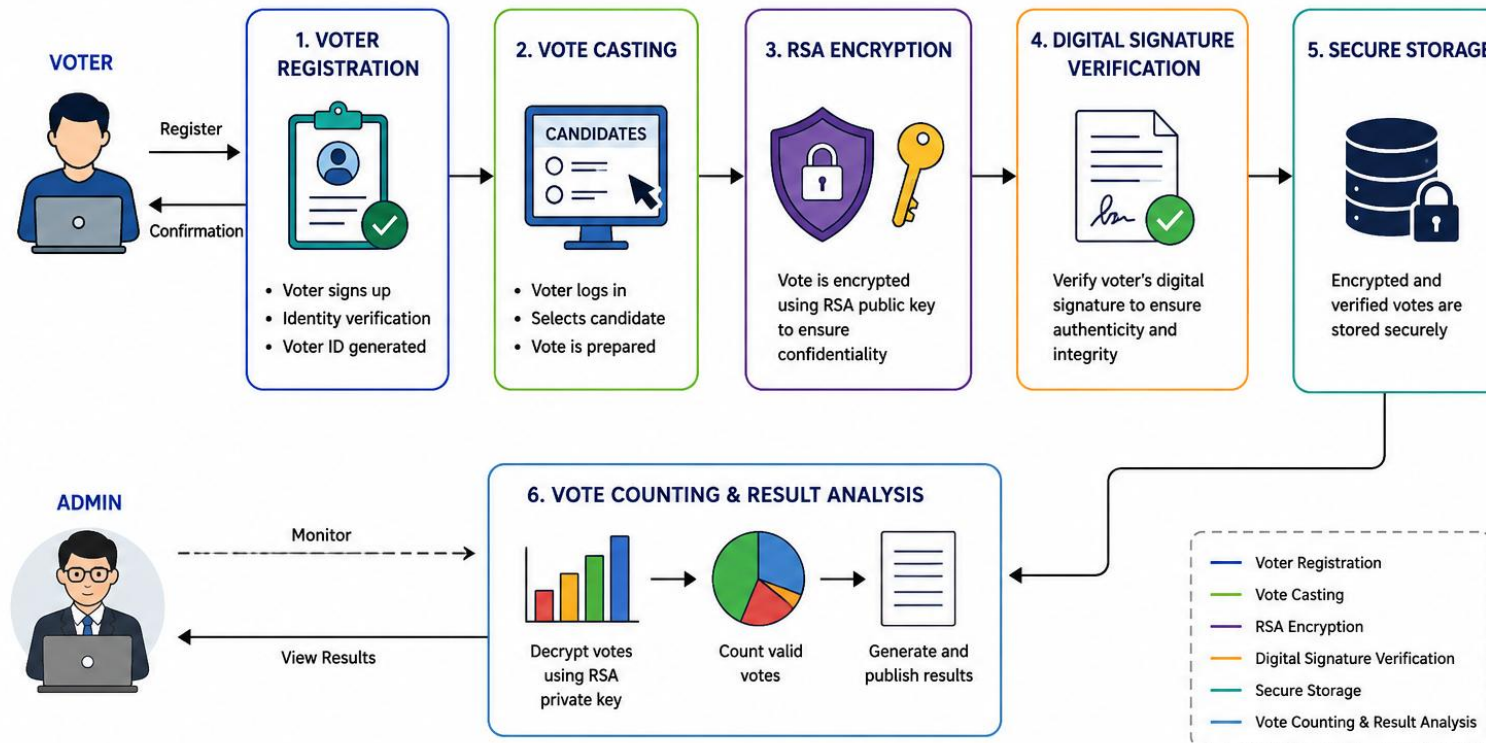
- The system provides a secure platform for online voting.
- Voters must register before participating in the election.
- Only authorized voters are allowed to cast their votes.
- Votes are protected using RSA-based encryption.
- Digital signatures are verified to ensure vote authenticity.
- Valid votes are securely counted to generate the final results.





ARCHITECTURE DIAGRAM

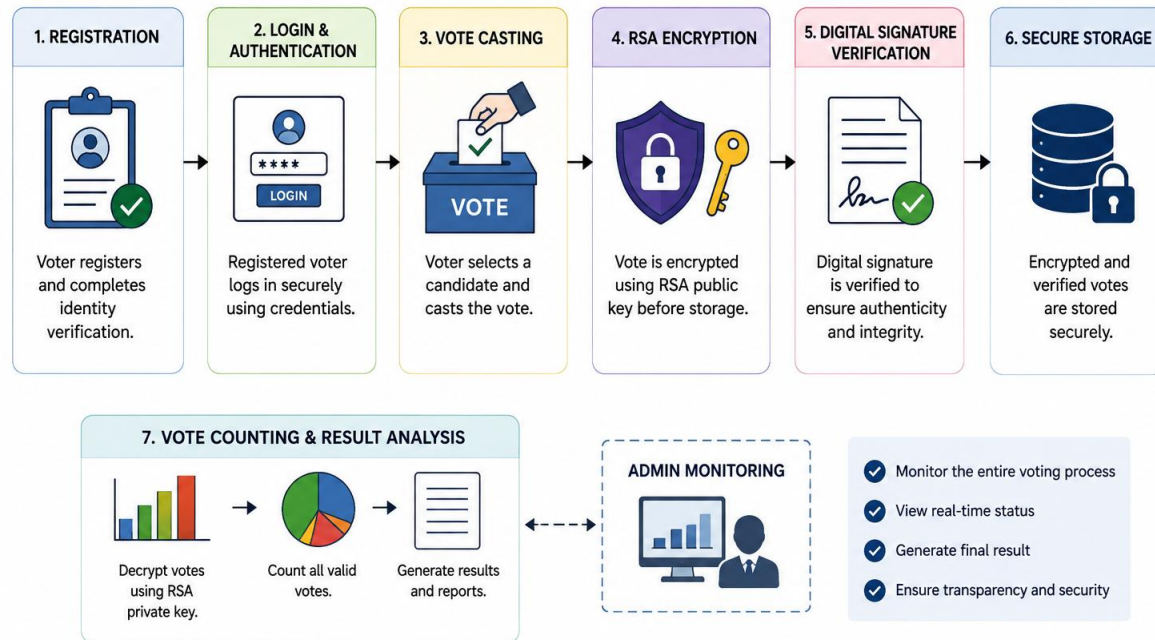
SECURE ONLINE VOTING SYSTEM ARCHITECTURE DIAGRAM





SYSTEM WORKFLOW

SYSTEM WORKFLOW



- The voter registers in the system.
- The voter logs in using valid credentials.
- The voter selects a candidate and casts the vote.
- The vote is encrypted using RSA encryption.
- The digital signature is verified for authenticity.
- The encrypted vote is stored securely.
- Valid votes are counted and results are generated.



MODULE 1 :VOTER REGISTRATION



- Allows eligible voters to register in the system.
- Collects basic voter details and login credentials.
- Creates a unique voter ID for each voter.
- Verifies the voter before allowing access.
- Prevents duplicate voter registration.
- Ensures that only authorized voters can vote.





MODULE 2 :RSA-BASED VOTE ENCRYPTION



- The voter selects a candidate and casts the vote.
- The vote is converted into digital data.
- RSA encryption is applied to the vote.
- The public key is used to encrypt the vote.
- The encrypted vote cannot be read directly by others.
- This ensures vote confidentiality and security.



MODULE 3: DIGITAL SIGNATURE VERIFICATION



- Digital signatures verify the identity of the voter.
- Each vote is checked for authenticity.
- The system verifies that the vote is not modified.
- Invalid or tampered votes are rejected.
- This ensures vote integrity and authenticity.
- Only verified votes are accepted for counting.





MODULE 4 : VOTE COUNTING & RESULT ANALYSIS



- Only valid and verified votes are considered.
- Encrypted votes are securely processed for counting.
- Votes are counted for each candidate.
- The system calculates the total votes received.
- Results are generated automatically.
- The candidate with the highest votes is declared the winner.



TOOLS & TECHNOLOGIES USED



- **Python** – Backend development
- **Flask** – Web application framework
- **HTML** – Web page structure
- **CSS** – User interface design
- **JavaScript** – User interaction
- **SQLite** – Database management
- **RSA Algorithm** – Vote encryption
- **Digital Signature** – Authentication and verification



SECURITY FEATURES



- Only registered voters can access the system.
- Each voter is allowed to vote only once.
- RSA encryption protects vote confidentiality.
- Digital signatures verify voter authenticity.
- Vote tampering can be detected and prevented.
- Encrypted votes are stored securely.
- Only valid votes are considered for counting.





RESULTS

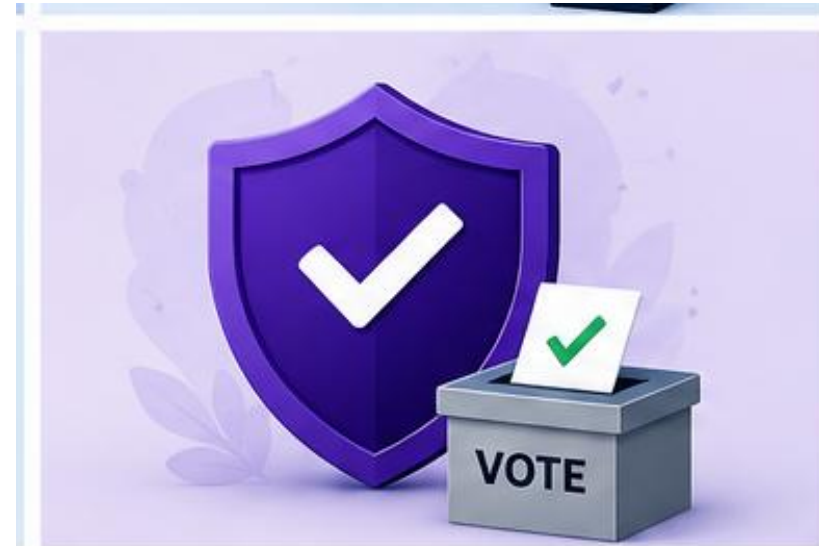


- Voters can register and access the system securely.
- Only authorized voters can cast their votes.
- Votes are successfully protected using RSA encryption.
- Digital signatures verify the authenticity of votes.
- Duplicate and invalid votes are prevented.
- Valid votes are counted accurately.
- The system generates secure and reliable election results.



CONCLUSION

- The system provides a secure and reliable online voting process.
- Voter registration ensures only authorized users can vote.
- RSA encryption protects the confidentiality of votes.
- Digital signatures ensure vote authenticity and integrity.
- Duplicate and unauthorized voting can be prevented.
- The system securely counts votes and generates accurate results.
- Overall, cryptography improves the security and reliability of online voting.





FUTURE ENHANCEMENTS



- Add biometric authentication for voters.
- Implement OTP-based verification.
- Use blockchain for more secure vote storage.
- Develop a mobile application for voting.
- Add multi-factor authentication.
- Improve scalability for large-scale elections.
- Add real-time security monitoring.



REFERENCES

1. Thakur, G., Chouksey, P., Chopra, M., & Sadotra, P. (2025). *Fortifying e-voting systems: Integrating visual cryptography with ECC and ChaCha20-Poly1305 for enhanced security*. **Journal of Communications Software and Systems**, **21**(4), 427–435. <https://doi.org/10.24138/jcomss-2025-0135>
2. Kraavi, T., & Willemson, J. (2025). *Proving vote correctness in the IVXV internet voting system*. **Scientific Reports**, **15**, Article 31793. <https://doi.org/10.1038/s41598-025-16764-1>
3. Kho, Y.-X., Heng, S.-H., Tan, S.-Y., & Chin, J.-J. (2025). *A provably secure coercion-resistant e-voting scheme with confidentiality, anonymity, unforgeability, and CAI verifiability*. **PLOS ONE**. <https://doi.org/10.1371/journal.pone.0324182>
4. Rahman, T. A., Muztahid, M., Nobo, D. K. R., & Hakim, S. A. (2025). *AmarVote: A web-based ElectionGuard system with post-quantum guardian key protection and blockchain auditing*. In **Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security** (pp. 115–119). <https://doi.org/10.1145/3777555.3777570>
5. Alsuwat, E. (2026). *Secure electronic voting system based on blockchain with Dilithium authentication and zero-knowledge proof verification*. **Scientific Reports**. <https://doi.org/10.1038/s41598-026-62925-1>

Thank
you